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The effects of neurotoxins on web-geometry and web-building behaviour in *Araneus diadematus* Cl.

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Abstract

The process of orb weaving and the resultant orb web constitute a good example of a complex behavioural pattern that is still governed by a relatively simple set of rules. We used the orb spider *Araneus diadematus* as a model organism to study the effect of the three neurotoxins (scopolamine, amphetamine, and caffeine) on the spider's behaviour. Scopolamine was given at two concentrations, with the lower one showing no effects but the higher one reducing web-building frequency; there also appeared to be a weak effect on web geometry. Amphetamine and caffeine, on the other hand, both resulted in significant changes in both building frequency and web geometry, compared to the controls. Amphetamine webs retained their size but showed an increase in spiral spacing and radius irregularity, as well as a decrease in building efficiency. Caffeine led to a general decrease in size and a slight increase in spiral spacing, as well as radius irregularity. Furthermore, caffeine caused webs to be rounder. Our observations suggest that these neurotoxins disturb different parts of the web-building programme presumably by affecting different actions in the spider's CNS.
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Keywords: Invertebrate behaviour; Web building; Orb geometry; *Araneus diadematus* spider; Scopolamine; Amphetamine; Caffeine

1. Introduction

In recent years there has been a growing interest in the field of neuroethology, especially of the invertebrates, whose relatively simple brains and stereotyped behaviour facilitate studies aimed at coupling specific behavioural patterns to distinct neurobiological actions [1]. Using neuroactive drugs to inhibit certain behaviours and faculties can help uncover the underlying mechanisms governing the measurable behaviour. As our knowledge of both neurobiology and the neurophysiological actions of drugs increase, so do the requirements for analysing the behaviour in depth and quantitatively.

A promising model organism for neuroethological research is the orb weaver *Araneus diadematus*. It builds a two-dimensional orb web, which can be viewed as a truthful record of a complex behaviour pattern, 'frozen in time' as it were without the use of filming the behaviour itself [2–4]. After the spider has located a suitable web site, it builds a web, which is species-specific and always has the following and rather distinct stages: (i) creation of the hub and the first radii; (ii) completion of the frame, hub, and radii; (iii) construction of the auxiliary spiral; and (iv) construction of the capture spiral [5]. During the completion of these stages, the spider orientates using a number of sensory cues, such as previously laid threads [6,7], gravity [8,9], and idiothetic memory [2,10]. Although the web-building behaviour is rather complex, computer simulations have revealed that good approximations of real webs can be accomplished by virtual spiders, which follow only a few simple rules [11,12]. The resultant orb web is a complex geometrical structure, which, however, is easily photographed and analysed for a multitude of components. Furthermore, the geometry of the orb web has proven to be highly sensitive to internal and external factors influencing the spider; these include missing and regenerated legs [13–15], age and size [16–18], nutritional state [19,20], ambient temperature and wind [21], and drug intake [22].

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Because of the orb weaver's sensitivity to drugs and the ease of analysing the web, Hans Peters and Peter Witt, about 50 years ago, had great hopes to be able to develop a 'web test' [23], for example, to identify even very small quantities of CNS active drugs in the body fluids from humans. From the late 1940s until the mid 1960s, a large range of drugs were fed to spiders and screened for any effect on the orb webs' geometry (see Witt and Reed [22] for a review). Nearly all drugs tested had some effect, and although the web test proved not to be analytical enough for practical use, these drug experiments produced first valuable insight into the function of a complex behavioural pattern [23]. Since the seminal work by Peters, Witt, et al., our analytical tools have improved; specifically, we developed in our laboratory faster and better methods to record and analyse final web geometry [24], as well as the underlying building behaviour [11] under normal as well as 'drugged' conditions [25]. To test our tools, we elected to repeat earlier studies on scopolamine, methamphetamine (amphetamine), and caffeine. In the present study, the effect of three neurotoxins on both web-building geometries of the web was investigated using the morphometrics of the finished web structure, as well as a detailed analysis of the building behaviour. Thus, in effect, we could analyse both anatomy and ontogeny to use a morphological analogue. Thus, integrating behaviour into a web drug study was a novel development and made possible through recent advances in spider tracking [26,27].

The three neurotoxins used in this study are known to affect the geometry of the orb web [22]. The three drugs were chosen on the basis of their expected differences in targeting specific senses in the spider. The anticholinergic alkaloid scopolamine is postulated to disrupt the orientation ability in spiders [28], and amphetamine is assumed to disrupt the sensory control [29]. The last drug, caffeine, is known to affect memory and learning in *Drosophila melanogaster* [30,31]. This study was limited to an analysis of web-building behaviour and orb geometry, but the observed effects of the drugs were attempted coupled with specific senses and neurophysiological processes building on existing knowledge.

2. Materials and methods

2.1. Experimental design

Juvenile *A. diadematus* spiders were caught at Mols in eastern Jutland, Denmark. Individual spiders were allowed a minimum of 1 week of acclimatisation to standard lab conditions (25 ± 2 °C, 45 ± 5% relative humidity, 16:8 h light–dark), before participating in the experiment. During this period, spiders were housed individually in $30 \times 30 \times 5$ cm Perspex frames, separated by thin Vaseline smeared plastic sheets, where they were watered daily and fed one fruit fly per web constructed. Webs were cut each day with a hot wire leaving an intact radius in the north and south quadrant, thereby collapsing the web into a single vertical thread, which was left in the frame for the spider to ingest. Only spiders that built three or more consecutive webs were included in the experiment.

Five different treatments were used in this experiment with different individual spiders assigned to each treatment. All drug solutions were prepared by dissolving the specific drug in 5.0 ml of distilled water. One gram of fruit sugar (d-Fructose—Sigma–Aldrich Chemie, Germany) was added to sweeten the solution to facilitate spider ingestion, except for the amphetamine solution where 1.5 g was needed before the spiders accepted the solution.

(1) Scopolamine low: 15 mg crystalline scopolamine hydrobromide trihydrate (Fluka Chemika, Switzerland) was dissolved.
(2) Scopolamine high: 75 mg crystalline scopolamine hydrobromide trihydrate (Fluka Chemika) was dissolved.
(3) Amphetamine: 6 mg crystalline methamphetamine hydrochloride (Sigma–Aldrich Chemie) was dissolved; note that 1.5 mg of fruit sugar was added to this solution.
(4) Caffeine: 25 mg crystalline caffeine (trimethylxanthine) was dissolved.
(5) Control: no drug was added to the solution.

Prior to drug application, the webs were placed in the opening of a black back-lit box and photographed with a Nikon Coolpix 995 digital camera. Spiders were then gently removed from the webs and weighed before being returned. When the spiders resettled in the hub, they were fed the drug solution by applying one or two drops, depending on body weight, to the mouth parts using a 5-μl Hamilton syringe. Drops of various sizes were administered, so that each spider received approximately 0.15 μl solution per milligram of body weight. After drug application, the spiders were allowed half an hour to ingest the drops before being reweighed. The spiders were then returned to the webs, which were cut before the frames were placed back under standard lab conditions. The 5-ml solutions consisted of 5 g water and 1 g sugar plus a small amount of drug, so by applying a density of 1.2 mg/μl, the weight gain of the spiders yields a reliable estimate of the volume of solution ingested. From this estimate, the amount of drug intake (μg/mg body weight) was calculated. The solutions were always administered in the afternoon, and in the following afternoon, the resulting experimental webs were photographed.

The behaviour of the spiders during web building was recorded for a subset of spiders for the treatments with amphetamine, caffeine, and control. The chosen spiders were transferred to the experimental setup: placed in a laboratory with room temperature where artificial light was present to ensure that the dark period was not longer than 8 h. In there, they built at least two webs before participating in the experiment. The setup consisted of two frames placed in

the opening of a box with a white background and illuminated from the sides and above by three fluorescent tubes. The tubes emitted only red light, which is invisible for spiders, to allow recording at night. Two black and white surveillance cameras were fitted in front of the frames and connected to two VP112 scanning units (or Baker boxes), which were directly connected to a Macintosh Performa 460 computer. The Baker box scanned the image recorded by the surveillance camera to continuously monitor the position of the spider (for details, see Zschokke [32]). Every new position was transferred to the computer running the application Move Recorder for the Macintosh [32], which stored the positions as x–y coordinates. Spiders were allowed several days to acclimate to the experimental setup and were only used in experiments when they had built a minimum of three webs each. The spiders were administered the drugs using the method described above, after which they were returned to the setup. Additionally, the completed webs were photographed using the Coolpix 995 digital camera immediately after the recording session was terminated.

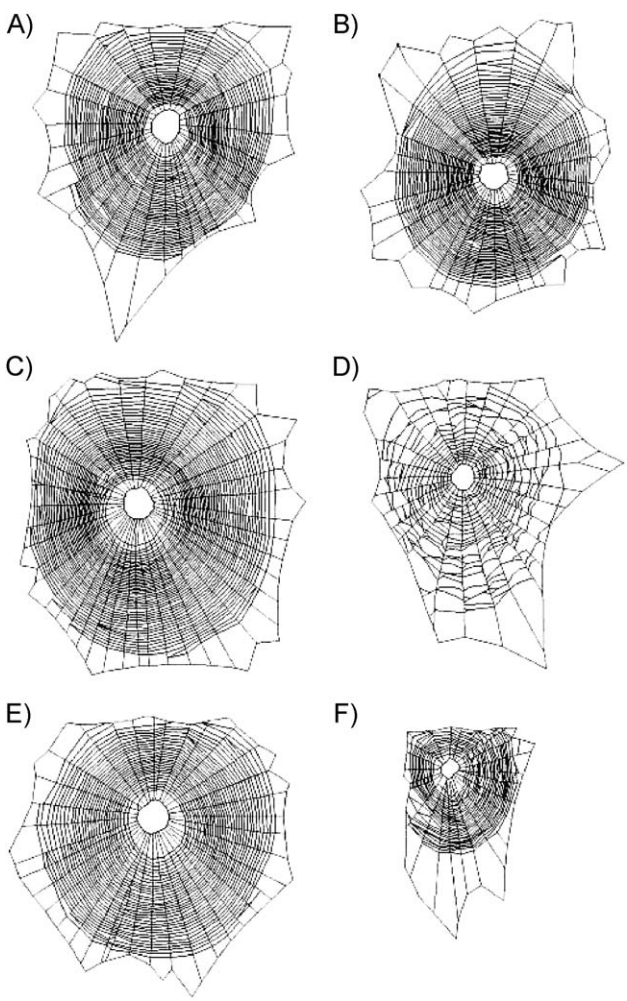

Fig. 1. Digitised photographs of webs from the orb weaver *A. diadematus.* The first column consists of webs made before the spiders were administered a drug solution, and the second column are webs from the same spiders made the day after the drug application. First row shows webs made before (A) and after (B) administration of a control solution. Second row shows webs made before (C) and after (D) administration of amphetamine. Third row shows web made before (E) and after (F) administration of caffeine.

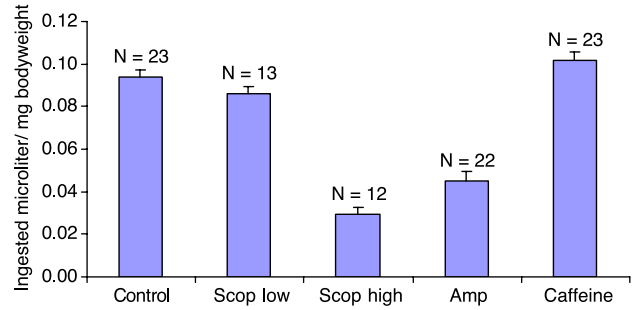

Fig. 2. Amount of solution ingested by the spider *A. diadematus*. Control refers to a control solution. Scop low and Scop high refer, respectively, to a low and a high concentration of scopolamine. Amp refers to amphethamine, and Caf refers to caffeine. The sample size for each solution is given above the bar. Error bars indicate the standard error of the mean.

2.2. Data analysis

The digital images were converted to pixel graphic and analysed using the software WebExtractor [33]. Additional measurements were obtained with a ruler from printouts of the images. From each web, the following data were extracted and analysed:

- Radii length—total length of all radii.
- Capture length—total lengths of all capture spiral threads.
- Capture area—the area covered by the capture spiral, excluding the hub and free zone.
- No. of reverses per centimeter of spiral thread—During capture spiral construction, the spider reverses its direction of thread laying several times, most often in

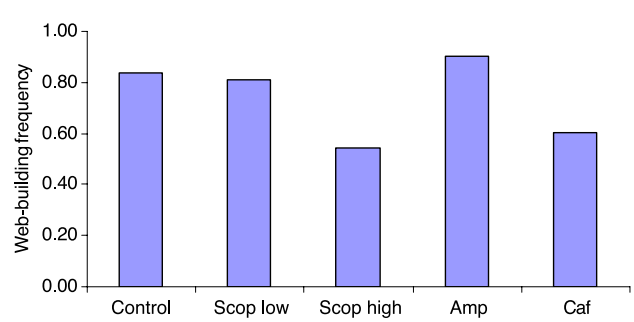

Fig. 3. Web-building frequency of *A. diadematus* the day following administration of drugs. For control (Control), 26 out of 31 spiders built a web. For the low concentration of scopolamine (Scop low), 13 out of 16 spiders built a web. For the high concentration of scopolamine (Scop high), 12 out of 22 spiders built a web. For amphetamine (Amp), 27 out of 30 spiders built a web. For caffeine (Caf) 23 out of 38 spiders built a web.

Table 1
The effects of administration of a control solution on web geometry

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α *
Number of spiders		23					
Weight (mg)	35.45 ± 2.17	<	39.56 ± 2.50				
Radii length (cm)	284 ± 14	>	279 ± 20	$t_{22} = 0.31$[a]	.756	>	.017
Capture spiral length (cm)	1083 ± 72	>	1033 ± 80	$t_{22} = 0.73$[a]	.474	>	.010
Capture area (cm^2)	239 ± 16	>	227 ± 17	$t_{22} = 0.81$[a]	.425	>	.008
Reverses per meter of spiral thread	0.64 ± 0.07	<	0.68 ± 0.09	$t_{22} = -0.43$[a]	.669	>	.013
Radial irregularity (%)	25.2 ± 1.8	<	25.3 ± 2.4	$t_{22} = -0.03$[a]	.976	>	.050
Mesh size (mm)	2.33 ± 0.07	<	2.34 ± 0.10	$t_{22} = -0.15$[a]	.881	>	.025
Eccentricity	0.47 ± 0.04	>	0.39 ± 0.05	$t_{22} = 1.69$[a]	.106	>	.007

All values are given with the S.E.M.
[a] Two-tailed paired t test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

the southern quadrant where more spiral turns are laid. A reverse was identified when two spiral threads joined together in a V shape. Inasmuch as the number of reverses presumably depends on the size of the web, it was here divided by capture length.

- Proportion of irregular radii—The number of Y-shaped (radii that splits into two inside the capture area) and distorted radii as a percentage of total number of radii. A radius was defined as distorted if it deviated by more than approximately 5° from a straight line as estimated visually from the printouts.
- Mesh size—Average mesh size was calculated by inserting a vertical line through the hub centre and employing the following formula derived by Herberstein and Tso [34].

$$\frac{1}{2}\left(\frac{r_u - \mathrm{Hr}_u}{S_u - 1} + \frac{r_l - \mathrm{Hr}_l}{S_l - 1}\right)$$

Where r_u and r_l are the upper and lower vertical capture spiral lengths, and Hr_u and Hr_l are the upper and lower vertical lengths of the hub and the free zone (the area between the hub and the capture spiral) taken together. In this study, the centre of the web was defined as the midpoint of the vertical line through the free zone and the hub, thereby keeping Hr_u and Hr_l of equal lengths. S_u and S_l are the numbers of capture spiral turns in the upper and lower part, respectively.

- Eccentricity—a measure of the shape of the web, calculated by using the following formula [21].

$$\sqrt{1 - \left(\frac{L_{hor}}{L_{vert}}\right)^2}$$

Where L_{hor} is the width, and L_{vert} is the height of the capture spiral. If the width is longer than the height, the term in the brackets is inverted and eccentricity is assigned a negative value.

From the recordings of movements during web building, the following measurements were obtained and analysed:

- Total building time—the total time elapsed from the spider creating the first radii until finishing the last capture spiral. Pauses during which the spider sits motionless were included in this time.
- Pauses—capture spiral. Number of times the spider was motionless longer than 20 s during construction of the capture spiral.

Table 2
The effects of administration of a low concentration of scopolamine on web geometry

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α *
Number of spiders		13					
Weight (mg)	28.53 ± 1.31	<	31.48 ± 1.48				
Drug intake (μg/mg)	–		0.26 ± 0.01				
Radii length (cm)	259 ± 25	>	245 ± 20	$t_{12} = 1.12$[a]	.287	>	.010
Capture spiral length (cm)	932 ± 90	>	846 ± 79	$t_{12} = 1.57$[a]	.143	>	.007
Capture area (cm^2)	310 ± 31	>	303 ± 29	$t_{12} = 0.44$[a]	.667	>	.025
Reverses per meter of spiral thread	0.68 ± 0.11	>	0.65 ± 0.09	$t_{12} = 0.38$[a]	.709	>	.050
Radial irregularity (%)	7.9 ± 1.5	<	10.9 ± 3.5	$t_{12} = -0.68$[a]	.507	>	.017
Mesh size (mm)	2.36 ± 0.12	<	2.54 ± 0.13	$t_{12} = -1.48$[a]	.164	>	.008
Eccentricity	0.57 ± 0.02	>	0.50 ± 0.07	$t_{12} = 1.07$[a]	.307	>	.013

All values are given with the S.E.M.
[a] Two-tailed paired t test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

Table 3
The effects of administration of a high concentration of scopolamine on web geometry

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α*
Number of spiders		12					
Weight (mg)	36.45 ± 2.26	<	37.75 ± 2.39				
Drug intake (μg/mg)	–		0.44 ± 0.05				
Radii length (cm)	247 ± 21	>	207 ± 17	$t_{11} = 1.71^a$	.114	>	.010
Capture spiral length (cm)	1015 ± 99	>	801 ± 68	$t_{11} = 2.30^a$	.042	>	.007
Capture area (cm^2)	315 ± 27	>	264 ± 25	$t_{11} = 1.62^a$	.134	>	.013
Reverses per meter of spiral thread	0.72 ± 0.12	<	0.75 ± 0.08	$t_{11} = -0.19^a$	.851	>	.050
Radial irregularity (%)	12.3 ± 1.6	<	24.5 ± 5.9	$t_{11} = -2.19^a$	.051	>	.008
Mesh size (mm)	2.41 ± 0.16	>	2.39 ± 0.16	$t_{11} = 0.21^a$	.838	>	.025
Eccentricity	0.57 ± 0.06	<	0.61 ± 0.02	$Z = -0.51^b$	.610	>	.017

All values are given with the S.E.M.
[a] Two-tailed paired t test.
[b] Wilcoxon's signed-rank test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

- Speed (auxiliary spiral)—the average building speed of the spider when constructing the auxiliary spiral. Pauses were not included in this measurement.
- Speed (capture spiral)—the average building speed of the spider when constructing the capture spiral. Pauses were not included in this measurement.
- Pause from aux to cap—the pause between finishing the auxiliary spiral and starting on the first capture spiral turn. This pause is seen in all orb weavers [5] and could be a necessary break used to change from normal silk to gluey silk.
- Building efficiency—the efficiency exhibited by the spider during capture spiral construction. It was calculated by dividing the length of all capture spiral threads with the distance the spider had moved during capture spiral construction.

Statistical comparisons between the effects of each drug and the control, on amount of solution ingested and web-building frequency, were made using a two-tailed Student t test for independent samples and a chi-square test with the Yates correction. For all other measurements, comparisons were made between the predrug webs and the postdrug webs for each individual spider and paired t tests conducted if possible. In some cases, even transformations did not result in equal variances; here, the nonparametric Wilcoxon signed-rank test for matched samples was used. The radial irregularity was subjected to the arcsine transformation before a paired t test was attempted. Inasmuch as several tests were performed on the same data set, a sequential Bonferroni adjustment was made to minimise type-I errors [35]. The initial significance level was set at 5%. All statistical tests were performed using the software SPSS 10.0 (SPSS 1999).

3. Results

Before the chosen parameters of the webs are analysed and compared between pre- and postdrug webs below, it is important to remember that the collected data consisted of photographs. A visual-based comparison between webs made before and after administration of the solution was made. This was based on ranking all photographs in the three categories: size, shape, and web regularity. For the control solution, the comparison revealed, as expected, a

Table 4
The effects of administration of amphetamine on web geometry

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α*
Number of spiders		22					
Weight (mg)	33.07 ± 1.78	<	34.99 ± 1.89				
Drug intake (μg/mg)	–		0.06 ± 0.01				
Radii length (cm)	302 ± 16	>	297 ± 13	$t_{21} = 0.25^a$	.809	>	.05
Capture spiral length (cm)	1174 ± 70	>	752 ± 59	$t_{21} = 4.53^a$	1×10^{-4}	<	.008
Capture area (cm^2)	254 ± 17	>	241 ± 14	$t_{21} = 0.66^a$	.514	>	.025
Reverses per meter of spiral thread	0.63 ± 0.08	>	0.50 ± 0.07	$t_{21} = 1.45^a$	.161	>	.013
Radial irregularity (%)	20.0 ± 1.8	<	64.4 ± 6.2	$Z = -3.69^b$	2×10^{-4}	<	.010
Mesh size (mm)	2.24 ± 0.07	<	4.26 ± 0.46	$Z = -4.11^b$	4×10^{-5}	<	.007
Eccentricity	0.50 ± 0.04	>	0.44 ± 0.08	$Z = -0.66^b$	.509	>	.017

All values are given with the S.E.M.
[a] Two-tailed paired t test.
[b] Wilcoxon's signed-rank test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

Table 5
The effects of caffeine on web geometry

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α*
Number of spiders		23					
Weight (mg)	37.01 ± 7.88	<	41.59 ± 9.11				
Drug intake (μg/mg)	–		0.26 ± 0.01				
Radii length (cm)	293 ± 21	>	188 ± 17	$t_{22}=6.67^{a}$	1×10^{-6}	<	.010
Capture spiral length (cm)	1212 ± 110	>	537 ± 79	$t_{22}=7.98^{a}$	6×10^{-8}	<	.007
Capture area (cm^2)	247 ± 20	>	121 ± 14	$t_{22}=7.76^{a}$	1×10^{-7}	<	.008
Reverses per meter of spiral thread	0.67 ± 0.08	<	0.84 ± 0.09	$t_{22}=-1.68^{a}$	.108	>	.050
Radial irregularity (%)	23.6 ± 1.9	<	40.0 ± 2.9	$t_{22}=-5.57^{a}$	1×10^{-5}	<	.013
Mesh size (mm)	2.19 ± 0.08	<	2.72 ± 0.20	$Z=-2.24^{b}$	.025	>	.025
Eccentricity	0.49 ± 0.04	>	0.21 ± 0.09	$Z=-2.69^{b}$	.007	<	.017

All values are given with the S.E.M.
[a] Two-tailed paired t test.
[b] Wilcoxon's signed-rank test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

high degree of similarity (see Fig. 1A,B). However, slight variations in size and especially shape were common, and for 6 out of 23 pairs, the difference was large enough to be obvious even with a quick visual scan eyeballing of the two photographs. For the low and high concentrations of scopolamine, 3 out of 13 and 5 out of 12, respectively, showed obvious visual differences. The amphetamine postdrug webs were very characteristically with a large mesh size and slack capture threads especially towards the periphery (see Fig. 1D). However, 7 out of 22 lacked these differences and were quite similar to the predrug webs. For caffeine, the postdrug webs were again characteristically with a clear reduction in size (see Fig. 1F); and here, only 3 out of 23 postdrug webs were similar to the predrug webs.

In Fig. 2, the actual amount of drug solution ingested by the spiders for the different treatments is shown. Inasmuch as approximately 0.15 μl/mg bodyweight drug solution was given to each spider in each treatment, it can be seen from Fig. 2 that in no cases was all the administered solution ingested. Comparisons of amount ingested between the control and each of the drug solutions showed that administering a high concentration of scopolamine and amphetamine resulted in a significant decrease of solution ingested, whereas a low concentration of scopolamine and caffeine gave no significant difference in amount ingested compared to control (t test; low concentration of scopolamine: $t=1.54$, $df=34$, $P>.050$; high concentration of scopolamine: $t=12.23$, $df=33$, $P<.013$; amphetamine: $t=8.96$, $df=43$, $P<.017$; caffeine: $t=-1.61$, $df=44$, $P>.025$). This implies that amphetamine and scopolamine in high doses could be detected by the spider and presumably have an adverse taste. This could possibly have been masked in the experiment by adding more sugar to the solution, but trials with increased d-fructose concentration resulted in problems with completely dissolving the sugar and a very viscous solution.

The administered drugs gave rise to different web-building frequencies (see Fig. 3). A statistical analysis revealed that only the high concentration of scopolamine and caffeine differed significantly in web-building frequency compared to the control (χ^2 test; low concentration of scopolamine: $\chi^2=0.003$, $df=1$, $P>.050$; high concentration of scopolamine: $\chi^2=11.9$, $df=1$, $P<.017$; amphetamine: $\chi^2=0.44$, $df=1$, $P>.025$; caffeine: $\chi^2=13.6$, $df=1$, $P<.013$). Although not significant, the results indicate that amphetamine had actually brought about a higher web-building frequency than the control had.

The effects of the drugs on the web geometry are summarised in Tables 2–5. Table 1 shows that there were no significant effects on the webs after application of the

Table 6
The effect of administration of a control solution on web-building behaviour

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α*
Number of spiders		6					
Delay time (h)[a]	–		18.5 ± 1.8				
Total building time (min)	53.0 ± 6.0	>	40.5 ± 5.2	$t_5=1.53^{b}$	.186	>	.017
Number of pauses—capture	0		0				
Speed—auxiliary (mm/s)	7.1 ± 0.3	<	8.3 ± 0.7	$t_5=-2.00^{b}$	.101	>	.010
Speed—capture (mm/s)	5.4 ± 0.3	<	5.7 ± 0.5	$t_5=-0.95^{b}$	.385	>	.025
Pause from aux to cap (s)	28 ± 4	>	26 ± 3	$t_5=0.37^{b}$	.725	>	.050
Building efficiency (%)	95.3 ± 1.5	<	99.0 ± 1.5	$t_5=-1.64^{b}$	.161	>	.013

All values are given with the S.E.M.
[a] Delay time is the duration from drug administration until onset of web building.
[b] Two-tailed paired t test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

Table 7
The effect of administration of amphetamine on web-building behaviour

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α*
Number of spiders		8					
Drug intake (μg/mg)	–		0.06 ± 0.01				
Delay time (h)[a]	–		16.7 ± 0.8				
Total building time (min)	51.5 ± 9.0	<	76.1 ± 11.6	$t_7 = -2.44$[b]	.045	>	.025
Number of pauses—capture	0	<	2.3 ± 1.4				
Speed—auxiliary (mm/s)	11.0 ± 0.6	>	9.0 ± 0.4	$t_7 = 1.93$[b]	.005	<	.013
Speed—capture (mm/s)	6.6 ± 0.2	>	4.9 ± 0.3	$t_7 = 4.66$[b]	.002	<	.010
Pause from aux to cap (s)	18 ± 2	<	31 ± 10	$Z = -1.47$[c]	.141	>	.050
Building efficiency (%)	101.0 ± 1.5	>	71.8 ± 10.5	$Z = -2.38$[c]	.017	>	.017

All values are given with the S.E.M.
[a] Delay time is the duration from drug administration until onset of web building.
[b] Two-tailed paired t test.
[c] Wilcoxon's signed-rank test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

control solution. There therefore seems to be no effect on web geometry of 'sugar', 'treatment', or 'handling'. Administration of a low concentration of scopolamine did not result in any significant effects on any of the web parameters analysed (see Table 2), although a slight increase in the proportion of irregular radii could be detected. The high concentration of scopolamine (see Table 3), however, doubled the proportion of irregular radii, compared with the predrug webs, and resulted in a decrease in radii and capture spiral length and a reduction of capture spiral area. Amphetamine had severe effects on web geometry (see Table 4), resulting in a significant threefold rise of irregular radii. The length of all capture spiral threads was significantly reduced, whereas the area of the capture spiral was unchanged. This led to a significantly larger mesh size in the spiral. Application of caffeine also resulted in profound changes in web geometry (Table 5). Total length of all radii, the length of the capture spiral, and the area of the capture spiral were all significantly reduced. The proportion of irregular radii was almost twice as high compared to predrug webs. The eccentricity of the webs was significantly lower, resulting in more round webs. Although there was no significant change in the number of reverses, the data did indicate a rise. Mesh size was also slightly larger in the postdrug webs, although this was not statistical significant.

Table 6 shows that there were no effects of the control solution on web-building behaviour. The relatively large decrease in total web-building time observed was mostly the result of fewer pauses during radial and frame construction, which is also the stage in the web-building sequence known to exhibit highest variation [2]. A larger sample size would presumably remove this effect. Tables 7 and 8 summarise effects of amphetamine and caffeine on the behaviour of the web-building spider. Inasmuch as scopolamine showed no significant effects on web geometry, it was decided not to record the web-building behaviour of scopolamine application.

Note the unrealistically high capture spiral building efficiency observed throughout this study. One explanation for this is that the body of the spider, inasmuch as it is inside the capture spiral being built, has a shorter trajectory than the capture spiral laid. Alternatively, factors could be either an overestimation of capture spiral length made by the digitising software (WebExtractor) or that not all movements made during the construction of the capture spiral

Table 8
The effect of administration of caffeine on web-building behaviour

	Predrug treatment		Postdrug treatment	Test	P		Adjusted α*
Number of spiders		7					
Drug intake (μg/mg)	–		0.27 ± 0.01				
Delay time (h)[a]	–		16.5 ± 0.8				
Total building time (min)	48.3 ± 6.8	>	46.5 ± 6.0	$t_6 = 0.19$[b]	.859	>	.050
Number of pauses—capture	0		0				
Speed—auxiliary (mm/s)	8.8 ± 0.6	>	5.4 ± 1.0	$t_6 = 3.70$[b]	0.010	<	.013
Speed—capture (mm/s)	6.4 ± 0.4	>	4.5 ± 0.6	$t_6 = 4.10$[b]	.006	<	.010
Pause from aux to cap (s)	20 ± 1	<	64 ± 34	$Z = -2.20$[c]	.028	>	.017
Building efficiency (%)	103.7 ± 1.9	>	92.9 ± 5.7	$Z = -1.99$[c]	.046	>	.025

All values are given with the S.E.M.
[a] Delay time is the duration from drug administration until onset of web building.
[b] Two-tailed paired t test.
[c] Wilcoxon's signed-rank test.
* Inasmuch as several tests were performed on the same data set, the significance levels were adjusted using the sequential Bonferroni method.

were actually recorded by the Baker boxes. The application of both amphetamine and caffeine showed visual effects on the spiders within an hour, resulting in more sluggish and passive spiders. This immediate effect, however, did not result in delay in onset of web building, as the time from application of the drug solution to the onset of web building was roughly the same for the drugged and control spiders (see Tables 6, 7, and 8).

For amphetamine, the significant effects on web-building behaviour were a reduction in speed, both during auxiliary and capture spiral construction (see Table 7). There was also an apparent reduction in capture spiral building efficiency, although this was statistically just insignificant. The data also indicated an increase in total building time. Caffeine resulted in a reduction of speed during both auxiliary and spiral construction (see Table 8). However, contrary to amphetamine, caffeine led to a significant increase in the pause after finishing the auxiliary spiral and before commencing on the capture spiral. Caffeine, furthermore, did not result in changes in total building time or capture spiral building efficiency, although the latter was close to being statistically significant.

4. Discussion

Orb weavers exhibit a high degree of individual variation in their responses to biotic [36,37] as well as abiotic environmental factors [21,38]. It is therefore not surprising that *A. diadematus* showed individual variation in the effects of the neurotoxins examined here. Some individuals even built seemingly normal webs after ingestion of the same doses at which other individuals built highly distorted webs. Despite this, the results of the present study agree with earlier work on spiders' responses to drugs (see Witt and Reed [22] for a review) in that drug-specific actions were found on the geometry of the orb web. The experiments on scopolamine conducted in this study showed that web-building frequency was dose-dependent and indicated that the higher concentration and higher drug intake influenced several web parameters, whereas the lower concentration of scopolamine showed no such influence. Dose-dependent effects have been reported for a wide range of drugs [17,28,39]. The concentrations chosen in this study were, on one hand, a compromise between the wish to observe actual changes and, on the other, to ensure that the drug solution was readily ingested by the spider and that web-building frequency was high enough to ensure an appropriate sample size. It would have been interesting with even higher concentrations of scopolamine, but trial experiments conducted with higher concentrations resulted in active rejection of the drop attached to the mouth parts by the spider moving out of the hub and wiping off the drops on the ground. Samu and Vollrath [25] used a different approach to ensure ready acceptance of liquid pesticides sweetened with sugar. They deprived the spiders of water for 3–5 days prior to drug application. This approach was discarded in this study as it was feared that water deprivation would result in an increased variation in web parameters and that the sudden ingestion of much solution on the day of drug application could lead to differences between pre- and postadministration webs in the control experiment.

Administration of a low concentration of scopolamine resulted in an actual intake by the spiders of around 0.26 μg/mg spider body weight. This dose had no effect on either web-building frequency or any of the web parameters investigated. Raising the concentration of scopolamine fivefold resulted only in a rise of intake to 0.44 μg/mg due to a much lower amount of drug solution actually ingested by the spiders. However, this rise in dose was enough to significantly reduce web-building frequency, and although not statistically significant, there was an indication that it also reduced web size and led to a higher irregular radii. Earlier work on the effect of scopolamine was conducted on adult *Zygiella x-notata*. Wolff and Hempel [28] do not state the actual dose given per milligram of bodyweight, but it can be calculated using their average dose applied and dividing it with the mean weight of adult female *Z. x-notata*, giving a dose of around 0.7 μg/mg. Witt and Reed [22] mention a dose as low as 0.05 μg/mg; however, as their paper is a review (one of the papers reviewed was Wolff and Hempel's study mentioned above), this value might be a miscalculation. After administration of scopolamine, they found no change in web-building frequency, an increase in eccentricity, and a decrease in both size and regularity. The first two results seem to be contradicted by the findings of the present study, where there was a clear reduction in web-building frequency and no sign of an increase in eccentricity. However, this might be explained by the use of juvenile *A. diadematus* instead of adult *Z. x-notata*.

In humans, scopolamine is known to antagonise acetylcholine at the muscarinic receptors in the CNS [40], and it has negative effects on memory and learning in monkeys as well as in humans [41,40]. Similar effects are also reported in octopuses [42]. In spiders, acetylcholine is found in the CNS [43,44], and the enzyme acetylcholinesterase, which destroys acetylcholine, has been found in high concentrations in parts of the brain thought to be associated with motor control for orb weaving [45]. Scopolamine would therefore be expected to have a significant impact on spider behaviour and result in changes in orb web geometry. The relative weak effects observed in this study might be due to a too-low dose given. However, it is also possible that the method of oral application is not well suited for this particular drug. Larvae of the pest *Spodoptera littoralis* show no adverse effect when eating plant material containing high concentrations of scopolamine [46]. The authors speculate that this could be due to insufficient permeability of scopolamine through the gut membrane.

The concentration of amphetamine given in this experiment led to a drug intake of 0.06 μg/mg and resulted in a

decrease in capture spiral length but no change in radii length or capture spiral area. Furthermore, we observed an increase in radial irregularity and mesh size. An analysis of the movements made by the spider during web building showed a clear reduction in speed both during auxiliary and capture spiral construction compared to the control. Interestingly enough, there also seemed to be a reduction in capture spiral construction efficiency, albeit not significantly so, where the spider moved longer, compared to the control, for laying down the same length of spiral thread. Older studies on adult *A. diadematus* and *Z. x-notata* showed that a low dose of 0.1 μg/mg resulted in an increase in web-building frequency and web size, whereas a high dose of 0.4–1.0 μg/mg resulted in a decrease in web-building frequency, in the number of radii and spiral turns, in capture area and in regularity [22]. This agrees well with the results found in this study, although the juvenile *A. diadematus* used here seemed more sensitive to amphetamine. However, although this was not statistically significant, a higher web-building frequency was also observed in this study after administration of amphetamine, compared to the control. Amphetamine is reported to disrupt the sensory control of the spider, causing it sometimes to reach into empty air when using the previously laid capture spiral for construction of the next [29].

Mesh size is fairly constant in the orb web; however, the outer spirals often show an increase in mesh size—a phenomenon which has been linked to prey capture efficiency [24,47]. As previously laid capture spiral turns are used during construction [9], it is expected that webs built by spiders experiencing sensory problems would show the largest disruption towards the periphery. This is indeed what was observed in this study for webs made after administration of amphetamine. The increase in proportion of irregular radii and especially the decrease in capture spiral construction efficiency found in this study further indicate that amphetamine-drugged spiders were experiencing problems with integrating sensory inputs.

In humans, amphetamine primarily destroys the dopaminergic and serotonergic terminals in the CNS, thereby causing release of the neurotransmitters dopamine and serotonin [48,49]. Dopamine and serotonin are both found in the spider CNS [50]. Serotonin seems to exert many different actions, but its main function is probably to transfer information from the afferent of the mechano- and chemoreceptive sensory hairs [51]. To our knowledge, there is no available information on the role of dopamine in the CNS of spiders yet; however, it is known to increase motivation in mammals [52]. A similar effect in spiders may be responsible for the indicated increase in web-building frequency after administration of amphetamine, as found in this study and reported by Witt and Reed [22].

The caffeine solution used here resulted in an average dose of 0.26 μg/mg. This caused a decrease in radii length, capture spiral length, and capture spiral area. There was also a slight increase in proportion of irregular radii as well as mesh size, although not as much as that observed for amphetamine. Webs made by caffeine-drugged spiders were characteristically more round than control webs. Furthermore, caffeine caused a reduction in speed during both auxiliary and capture spiral construction. It is noteworthy that caffeine seemed to prolong the natural pause of the spider between finishing the auxiliary spiral and beginning the capture spiral. Caffeine is claimed to be one of the most potent of the drugs tested on orb weavers [3,22,29]. These authors found that a dose of 1.0 μg/mg fed to adult female *Z. x-notata* caused a decrease in capture spiral area and led to more round webs and higher radial irregularity. Furthermore, they observed an increase in web-building time. These observations are in good agreement with the findings in the present study, although Witt et al. found more profound effects of caffeine on web geometry. This may be due to their use of a dose, which was more than threefold higher. However, unpublished trial experiments by us applying a caffeine dose of 0.42 μg/mg spider weight resulted in a web-building frequency of virtually zero and the death of half of the drugged spiders within a week.

Caffeine antagonises adenosine receptors in the human CNS [53]. To our knowledge, there is no published information about the role of adenosine in the spider CNS, but adenosine diphosphate (ADP) and adenosine triphosphate (ATP) play a fundamental role in the energetics of all living organisms. However, caffeine exerts different actions in man and spider. Caffeine has the effect, among others, on humans that it improves motor performance [53]; whereas on spiders, as the present study suggests, it seems to act the opposite by reducing the web-building speed and leading to structure disruption.

The experiments described in this study lend support to the hypothesis that neurotoxins affect the web geometry and web-building behaviour of *A. diadematus* in drug-specific ways. Our results are inconclusive as to whether scopolamine has the significant effects claimed by Witt et al., but they clearly supported the thesis of this research group that amphetamine and caffeine have profound and distinct effects.

The present study did not allow for conclusions about the issue of which exact senses or faculties were targeted by the drugs, although they provide some support to the notion that amphetamine disrupts motor sensory control. Amphetamine seems to be the drug with the greatest potential. However, even such a powerful drug, which caused profound changes in the capture spiral, did not cause any significant changes in the auxiliary spiral. This corresponds well with observations by Vollrath [8,9] that both spirals have totally different underlying construction rules [54]. Our recording of the movements made during auxiliary spiral constructions revealed, besides a reduction of construction speed, no visible changes to the path of the auxiliary spiral. It appeared to be just as regular after intake of amphetamine as before. Inasmuch as the auxiliary spiral is used as a handrail during construction of the capture spiral [6,26], it

would be interesting to find a drug that did cause a disruption of the auxiliary spiral. Scopolamine is described as having such effects [28], however, the results of the present study give no support to that.

Further experiments on other parts of spider behaviour such as predatory behaviour, escape movements, etc. could provide valuable additional information about which senses might be affected by the neurotoxins. It might be another interesting approach to use radioactive-labelled drugs, which would allow the determination of the locality in the spider's CNS where the drugs are concentrated. Together with neurophysiological methods, this could yield insight into which parts of the brain are important for which steps in the web-building process. Studying the application of drugs to orb weavers and analysing the effect using both web morphology and building ontogeny could, if coupled to more intensive CNS studies, prove to be a valuable new model system to generate new insights into invertebrate neuroethology.

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References

[1] Simmons P, Young D. Nerve cells and animal behaviour. 2nd ed. Cambridge: Cambridge Univ. Press; 1999.
[2] Vollrath F. Analysis and interpretation of orb spider exploration and web-building behavior. Adv Study Behav 1992;21:147–99.
[3] Witt PN, Reed CF, Peakall DB. A spider's web: problems in regulatory biology. Berlin: Springer-Verlag; 1968.
[4] Peters HM. Über das Kreuzspinnennetz und seine probleme. Naturwissenschaften 1939;47:77–786.
[5] Foelix RF. Biology of spiders. 2nd ed. Oxford: Oxford Univ. Press; 1996.
[6] Zschokke S. The influence of the auxiliary spiral on the capture spiral in *Araneus diadematus* Clerk (Araneidae). Bull - Br Arachnol Soc 1993;9:169–73.
[7] Peters PJ. Orb-web construction: interaction of spider (*Araneus diadematus* Cl.) and thread configuration. Anim Behav 1970;18:478–84.
[8] Vollrath F. Spiral orientation of *Araneus diadematus* orb webs built during vertical rotation. J Comp Physiol, A 1988;162:413–9.
[9] Vollrath F. Gravity as an orientation guide during web-construction in the orb spider *Araneus diadematus* (Araneae, Araneidae). J Comp Physiol, A 1986;159:275–80.
[10] Vollrath F, Nørgaard T, Krieger M. Radius orientation in the cross spider *Araneus diadematus*. In: Toft S, Scharff N, editors. European arachnology 2000. Århus: Aarhus Univ. Press; 2002. p. 107–16.
[11] Krink T, Vollrath F. Analysing spider web-building behaviour with rule-based simulations and genetic algorithms. J Theor Biol 1997;185: 321–31.
[12] Gotts NM, Vollrath F. Artificial intelligence modelling of web-building in the garden cross spider. J Theor Biol 1991;152:485–511.
[13] Weissmann M, Vollrath F. The effect of leg loss on prey capture in *Nephila clavipes*. Bull - Br Arachnol Soc 1999;11:198–203.
[14] Vollrath F. Lyriform organs on regenerated legs of the garden gross spider. Bull - Br Arachnol Soc 1995;10:115–8.
[15] Vollrath F. Altered geometry of webs in spiders with regenerated legs. Nature 1987;328:247–8.
[16] Heiling AM, Herberstein ME. The role of experience in web-building spiders (Araneidae). Anim Cogn 1999;2:171–7.
[17] Christiansen A, Baum R, Witt PN. Changes in spider webs brought about by mescaline, psilocybin and an increase in body weight. J Pharmacol Exp Ther 1962;136:31–7.
[18] Mayer G. Untersuchungen über Herstellung und Struktur des Radnetzes von *Aranea diadema* und *Zilla x-notata* mit besonderer Berücksichtigung des unterscheides von Jugend- und Altersnetzen. Z Tierpsychol 1953;9:337–62.
[19] Herberstein ME, Abernethy KE, Backhouse K, Bradford H, de Crespigny FE, Luckock PR, et al. The effect of feeding history on prey capture behaviour in the orb-web spider *Argiope keyserlingi* Karsch (Araneae: Araneidae). Ethology 1998;104:565–71.
[20] Vollrath F, Samu F. The effect of starvation on web geometry in an orb-weaving spider. Bull - Br Arachnol Soc 1997;10:295–8.
[21] Vollrath F, Downes M, Krackow S. Design variability in web geometry of an orb-weaving spider. Physiol Behav 1997;62:735–43.
[22] Witt PN, Reed CF. Spider-web building. Science 1965;149:1190–7.
[23] Witt PN. Drugs alter web-building of spiders: a review and evaluation. Behav Sci 1971;16:98–112.
[24] ap Rhisiart A, Vollrath F. Design features of the orb web of the spider, *Araneus diadematus*. Behav Ecol 1994;5:280–7.
[25] Samu F, Vollrath F. Spider orb web as bioassay for pesticide side effects. Entomol Exp Appl 1992;62:117–24.
[26] Zschokke S, Vollrath F. Unfreezing the behaviour of two orb spiders. Physiol Behav 1995;58:1167–73.
[27] Zschokke S, Vollrath F. Web construction patterns in a range of orb weaving spiders. Eur J Entomol 1995;92:523–41.
[28] Wolff D, Hempel U. Versuche über die Beeinflussung des Netzbaues von *Zilla x-notata* durch Pervitin, Scopolamin und Strychnin. Z Vgl Physiol 1951;33:497–528.
[29] Peters HM, Witt PN, Wolff D. Die Beeinflussung des Netzbaues der Spinnen durch Neurotrope Substanzen. Z Vgl Physiol 1950;32: 29–45.
[30] Dudai Y, Buxbaum J, Corfas G, Orgad S, Segal D, Sher B, et al. Defective cAMP metabolism and defective memory in Drosophila. Acta Biochim Biophys Hung 1986;21:177–92.
[31] Folkers E, Spatz HC. Visual learning performance of *Drosophila melanogaster* is altered by neuropharmaca affecting phosphodiesterase and acetylcholine transmission. J Insect Physiol 1984;30:957–65.
[32] Zschokke S. Web construction behaviour of the orb-weaving spider *Araneus diadematus* Cl. PhD thesis. Faculty of Philosophy and Science, University of Basel; 1994.
[33] Pedersen PO. Web extractor: processing of spider web images. M.Sc. Thesis. Department of Computer Science, University of Aarhus; 1995.
[34] Herberstein ME, Tso I-M. Evaluation of formulae to estimate the capture area and mesh height of orb-webs (Araneoidea, Araneae). J Arachnol 2000;28:180–4.
[35] Rice WR. Analyzing tables of statistical tests. Evolution 1989;43: 223–5.
[36] Heiling AM, Herberstein ME. Interpretations of orb-web variability: a review of past and current ideas. In: Gajdos P, Pekár S, editors. Proceedings of the 18th European Colloquium of Arachnology, Stará Lesná, 1999, vol. 19(Suppl. 3). Ekológia; 2000. p. 97–106.
[37] Schneider J, Vollrath F. The effect of prey type on the geometry of the capture web of *Araneus diadematus*. Naturwissenschaften 1998;85: 391–4.
[38] Sherman PM. The orb-web: an energetic and behavioural estimator of a spider's dynamic foraging and reproductive strategies. Anim Behav 1994;48:19–34.
[39] Reed CF, Witt PN. Progressive disturbance of spider web geometry caused by two sedative drugs. Physiol Behav 1968;3:119–24.

[40] Kopelman MD. The cholinergic neurotransmitter system in human memory and dementia: a review. Q J Exp Psychol 1986;38: 535–73.
[41] Harder JA, Baker HF, Ridley RM. The role of the central cholinergic projections in cognition: implications of the effects of scopolamine on discrimination learning by monkeys. Brain Res Bull 1998;45: 319–26.
[42] Fiorito G, Agnisola C, d'Addio M, Valanzano A, Calamandrei G. Scopolamine impairs memory recall in *Octopus vulgaris*. Neurosci Lett 1998;253:87–90.
[43] Fabian R, Seyfarth E-A. Acetylcholine and histamine are transmitter candidates in identifiable mechanosensitive neurons of the spider *Cupiennius salei*: an immunocytochemical study. Cell Tissue Res 1997;287:413–23.
[44] Meyer W, Schlesinger C, Poehling HM, Ruge W. Comparative quantitative aspects of putative neurotransmitters in the central nervous system of spiders (Arachnida: Araneidae). Comp Biochem Physiol 1984;78C:357–62.
[45] Meyer W, Pospiech B. The distribution of acetylcholinesterase in the central nervous system of web-building spiders (Arachnida, Araneae). Histochemistry 1977;51:201–8.
[46] Krug E, Proksch P. Influence of dietary alkaloids on survival and growth of *Spodoptera littoralis*. Biochem Syst Ecol 1993;21: 749–56.
[47] Witt PN. Ein einfaches Prinzip zur Deutung einiger Proportionen im Spinnennetz. Behaviour 1952;4:172–89.
[48] Davidson C, Gow AJ, Lee TH, Ellinwood EH. Methamphetamine neurotoxicity: necrotic and apoptotic mechanisms and relevance to human abuse and treatment. Brain Res Rev 2001;36:1–22.
[49] Frost DO, Cadet J-L. Effects of methamphetamine-induced neurotoxicity on the development of neural circuitry: a hypothesis. Brain Res Rev 2000;34:103–18.
[50] Schmid A, Sperk G, Reither H. Quantitative determination of neuroactive substances in the CNS of the spider *Cupiennius salei* KEYS. Comp Biochem Physiol 1992;102C:447–50.
[51] Barth F. A spider's world: senses and behaviour. Berlin: Springer-Verlag; 2002.
[52] Bradford HF. Chemical neurobiology: an introduction to neurochemistry. New York: W. H. Freeman and Company; 1986.
[53] Sawynok J, Yaksh TL. Caffeine as an analgesic adjuvant: a review of pharmacology and mechanisms of action. Pharmacol Rev 1993;45: 43–85.
[54] Vollrath F, Mohren W. Spiral geometry of the garden spider's orb web. Naturwissenschaften 1985;72:666–7.